

CHM111 Practice Exam on chapter(s): 1, 2

1. It is a compound. $\therefore$ it is a pure substance formed when two or more different chemical elements bond together in a fixed ratio.
2. He
3. 266.2 K
4. $0.10 \frac{\text{g}}{\text{mL}}$
5. Physical property. The density of lead is a physical property; it can be measured without changing the chemical identity of lead.
6. 433000000 W
7. 0.098120 mL
8. Pure substances: A, C
9. $9.066 \times 10^0 \text{ g}$
10. 417.57 nm
11. Element
12. Extensive property. Particle count is an extensive property $\therefore$ the number of particles depends on the size of the sample.
13. 53. g
14. Iron (III) sulfide: Fe_2S_3
15. 8.3 mg
16. Random error $\therefore$ the measurements fluctuate unpredictably in both directions.
17. $6.6 \times 10^{-11} \text{ g}$
18. Potassium most commonly forms the cation K^+ .
19. The SI base unit for amount of substance is the mole (symbol: mol).
20. $1.5 \times 10^2 \text{ g}$
21. $8.20 \times 10^1 \mu\text{g}$
22. 3.8 mm
23. Zinc fluoride hexahydrate: $\text{ZnF}_2 \cdot 6\text{H}_2\text{O}$
24. Nickel
25. 2256.2 m
26. Deci (symbol: d)
27. 44.4 g
28. $6.8 \times 10^{-5} \text{ m}$

29. Homogeneous mixture $\because$ it is a combination of two or more substances with a uniform composition and appearance.
30. Yes, a new substance is formed $\because$ Magnesium oxide is formed from magnesium and oxygen.
31. $2. \frac{\text{mol}}{\text{L}}$
32. Chemical change $\because$ oxidation reactions form new brown-colored compounds.
33. $75\text{ }^{\circ}\text{C}$
34. $4.030 \times 10^0 \text{ kg}$
35. 216000000000 Hz
36. Sample A $\because \rho_A = 2.22$ and $\rho_B = 1.3$
37. 30.7 cm
38. $3.204 \times 10^0 \text{ km}$
39. $5.39 \times 10^1 \text{ fm}$
40. 7.2 mL